

H2GUARD · DESIGN DECISIONS REGISTER

H2Guard design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2guard		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions gathered from the review note and the decision records; budget treated as a value-engineering target

H2Guard design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes P1 to P11 (cup openings, disc ledge and epoxy, head and controller fixings, standoffs, push-in fitting, mounting plate, lid as front panel, glands, test port bracket, 200 mm fan sleeve with fan plate, valve bracket)	Accept; change any one	Accept all	Every made part and assembly step	HGD-DDR-003, Table 1
2	Head mounting: on the back wall above the bench (ports 110 mm from the apparatus's back edge, 260 mm from its centre) or hung from the ceiling directly over the apparatus	(a) wall mounting, offset limit set by test at TRL 4; (b) ceiling drop rod	(a)	Head position and step 4	HGD-DDR-003, A1
3	Fan motor inside the wall sleeve, in the exhaust stream	(a) accept; (b) a fan with its motor outside the air stream	(a)	Fan choice	HGD-DDR-003, A2
4	Oxygen depletion monitoring when inert gas cylinders share the room	Add an oxygen sensor; leave to the site	None yet	A third sensor and cable if added	HGD-DDR-001, O2
5	Where the set points and the inventory rule come from in each country of the first users	National codes; ISO 26142; the user's own rules	None yet	Trip board threshold, firmware set points	HGD-DDR-001, O3
6	How airflow is proven	Tachometer only (as built); add a differential pressure switch	None yet; a blocked duct is not detected with the tachometer alone	A pressure switch and tubing at the fan if added	HGD-DDR-001, O4
7	Low-cost solenoid valve or a certified gas valve	Low-cost brass valve (as built); certified gas valve at several times the price	None yet, pending the valve's hydrogen compatibility and seat leak rating	Valve and its bracket holes	HGD-DDR-001, O5
8	Whether a full-span bump test trips the system or a lower span gas is used	1 % hydrogen, tripping the system as a proof test (as planned); about 0.5 % without a trip	None until the cup delivery factor is measured	First checks and safety stop S4	Review note, 2026-09-25
9	A second interlock output (dry contact) for H2Bench's bench supply	Add a relay output; leave H2Bench to switch from the valve signal	None yet (interface point with H2Bench)	A relay and terminal on the mounting plate if added	Review note, 2026-09-25
10			Keep, with the caption		

#	Decision needed	Options	Recommendation	Affects in the build	Source
	Render-only layout of the head and valve beside the controller in the product renders	Keep the render layout with a caption; render the true room layout		None (renders only)	Review note, 2026-09-26, item 1
11	Clear window in the controller lid	Clear-lid IP65 box; opaque lid as now specified	Clear-lid box, added to the BOM at its next revision	Controller box choice	Review note, 2026-09-26, item 2
12	Fan, make-up grille, sounder, beacon and supply left out of the product renders	Accept; add them	Accept	None (renders only)	Review note, 2026-09-26, item 3

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The catalytic sensor's response time (t90), power and hydrogen output, from its datasheet	R4 assumes a t90 of 30 s; R1's 1 % LFL resolution needs the output	HGD-CAL-001, F3
2	The sensor can is about 20 mm across and 17 mm tall	The standoff length sets its 2.2 mm gap above the disc	HGD-DDR-003, P4
3	The head box has its lid on a 110 x 90 face	The ports, gland and wall holes assume it	HGD-DDR-003, P3
4	The controller box's boss spacing and corner fixing holes	They set the mounting plate holes and the wall screws	HGD-DDR-003, P6, P8
5	The fan's body fits a 200 mm sleeve (about 180 mm across or less) and has fixing holes in its inlet face	The fan plate holds it by those holes	HGD-DDR-003, P10
6	The fan's curve gives at least 300 m ³ /h against about 60 Pa	R7 rests on an assumed curve	HGD-CAL-001, G3
7	The valve has two tapped holes underneath, 36 mm apart, and is rated for hydrogen with a stated seat leak rate	The bracket holes; the leaking-seat fault in Table 4 of HGD-CAL-001	HGD-DDR-003, P11; HGD-DDR-001, O5

■ Value engineering

Value-engineering target: USD 265 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 289 (USD 24 over the target), from the 16-line bill of materials at indicative prices. Main cost drivers and savings worth trying:

- The largest lines are the exhaust fan with its grille, sleeve, shutter and hood (USD 55), the catalytic sensor (USD 35), the solenoid valve (USD 30) and the controller modules (USD 25). Together they are about half the cost.
- Making the design constructable added USD 25: the made brackets and plates (USD 14), the 200 mm wall sleeve (USD 5), standoffs, screws and epoxy (USD 4) and the cup's push-in fitting and tube clips (USD 2).

- Savings worth trying: a printed controller mounting plate in place of aluminium sheet (about USD 3); buying the fan as a kit with its own shutter and hood; one combined controller board at TRL 4 in place of separate bought modules; and, where the room already has mechanical extract at high level, using it in place of the kit fan and grille (USD 63), which would bring the kit to about USD 226.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D9: catalytic plus metal oxide sensors (certified sensor as the upgrade); set points 10 % and 25 % LFL; normally closed valve; continuous ventilation with boost; inventory rule (R9); key-switch reset and one head per room; a university teaching lab first; independent hardware trip; 24 V DC with a certified supply	Amish: "i accept all your recommendations, go with them across all repos."	HGD-DDR-001, HGD-DDR-002
2026-09-25	Budget figure USD 265, now treated as the value-engineering target	Amish, same instruction; on 2026-10-01: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	HGD-DDR-002
2026-09-25	Timed escalation: a warning held 5 min closes the valve and latches	Amish, same instruction	HGD-DDR-002
2026-09-25	Head placement rule: a head directly above each likely leak point; offset limit set by test at TRL 4	Amish, same instruction	HGD-DDR-002
2026-09-25	R13 scope: the fan and grille wall openings are builder's work	Amish, same instruction	HGD-DDR-002
2026-09-25	H2Bench to reflect the inventory rule	Amish, same instruction	HGD-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan; keep open decisions out of the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	HGD-DDR-003 (changes open for review, open decision 1)